# QUESTIONS AND SOLUTIONS

## SECTION A: 5 marks each

1. A goat is tied to one corner of a square plot of side 15m by a rope 7m long. Find how much area the goat can graze on the plot.

**Solution**

The goat can graze on a quarter of the area of a circle centered at the corner of the plot where the goat is tied, and has radius equal to the length of the rope. Hence, the required area,

$$A = \frac{90}{360}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 7 \times 7 = \frac{77}{2} = 38.5cm^2.$$

2. James loves leftovers. On Thursday, he ate $\frac{1}{4}$ of the Pizza leftover; on Friday, $\frac{1}{3}$ of what remained then; on Saturday, $\frac{1}{2}$ of what remained then; and on Sunday he finished it off. On what day did he eat more than on any other day?

**Solution**

Let $J$ be the amount of Pizza leftover before Thursday's dinner. James eats $\frac{1}{4}J$ that day and $\frac{3}{4}J$ remains. On Friday, he eats $\frac{1}{3}$ of $\frac{3}{4}J$ which is $\frac{1}{4}J$ again, and $\frac{3}{4}J - \frac{1}{4}J = \frac{1}{2}J$ remains. On Saturday, he eats $\frac{1}{2}$ of $\frac{1}{2}J$ which is $\frac{1}{4}J$ again. Here, $\frac{1}{2}J - \frac{1}{4}J = \frac{1}{4}J$ remains, which he ate on Sunday. Clearly, James eats the same amount each day.

3. Alice and her brother John are walking together at the same pace. Alice stops to talk to Laura while John continues on at the same pace. After conversing with Laura, Alice continues on at the same pace as before. She later passes John, who stopped to sit on a bench. How long has John been sitting on the bench when Alice passes by?

**Solution**

Alice completes two tasks (talking to Laura, then walking to the bench) in the same time that her brother John completes two tasks (walking to the bench, then sitting on it). Since the walking pace is the same for both, the times spent walking to the bench are the same. Hence, John has been sitting on the bench for the same amount of time Alice talked to Laura.

4. In the sum shown, each letter represents a digit from 1 to 9, inclusive. What is the value of $P + Q + R$ where

$$QR + PPP + PPP = 2023.$$

**Solution**

If the value of $P$ is less than 9, then $QR + PPP + PPP$ is at most $99 + 888 + 888 = 1875$. Since the given sum is 2023, then the value of $P$ cannot be less than 9 and thus must equal 9. When $P = 9$, we get $QR + 999 + 999 = 2023$, and so $QR = 2023 - 999 - 999 = 25$. Therefore $Q = 2, R = 5$, and so $P + Q + R = 9 + 2 + 5 = 16$.

5. Ellena's birthday is on a Wednesday and Bruce's birthday is 121 days after Ellena's. On what day of the week is Bruce's birthday?

**Solution**

Since Ellena's birthday is on a Wednesday, then any exact number of weeks after Ellena's birthday will also be a Wednesday. Therefore, exactly 17 weeks after Ellena's birthday is also a Wednesday. Since there are 7 days in each week, then $7 \times 17 = 119$ days after Ellena's birthday is a Wednesday. Since 119 days after Ellena's birthday is a Wednesday, then 121 days after Ellena's birthday is a Friday (since 2 days after Wednesday is Friday). Therefore, Bruce's birthday is on a Friday.

6. When 15 is added to a set of 10 distinct natural numbers, the median changes from 6 to 8. Find the median of the new set if 7 replaces 15.

**Solution**

Originally, the median is the average of the 5th and 6th numbers. And the 6th number must be 8 (because it is the new median when 15 is added). Now replacing 15 with 7 puts 7 in the middle location (the 6th position). Hence the median of the new set is 7.

7. The sum of the ages of a father and his two sons who are twins is 48. Ten years hence, his age will be twice the sum of the sons' ages. Find the present age of their father?

**Solution**

Let the age of the father be $m$ years.

Let the age of one of his children be $n$ years and age of other child be $p$ years. Since they are twins, it means $n = p$.

From the given information, $m + 2n = 48$ .....(1)

After 10 years,

$(m + 10) = 2[(n + 10) + (p + 10)] = 2[2n + 20] = 4n + 40.$ .....(2)

Substituting equation (2) into equation (1), gives

$$4n + 30 + 2n = 48 \Longrightarrow 6n = 18 \Longrightarrow n = 3.$$

The age of the father is $48 - 6 = 42$ years.

8. In a row of children, David is 7th from the left and Vianney is 9th from the right. When they interchange among themselves, David becomes 18th from the left. What will be Vianney's present position from the right?

**Solution**

**Before interchange,**

David is 7th from the left and Vianney is 9th from the right. .....(1)

**After interchange**,

David is 18th from the left and 9th from the right. .....(2)

Vianney is 7th from the left. .....(3)

From (2) we get that there are total of $(18 + 9 - 1) = 26$ children in the given row. This together with (3) implies that the after interchange, Vianney is $(26 - 7 + 1) = 20$th from the right.

9. In a garden, there are 14 horizontal lines and 16 vertical lines of avocado trees. The distance between any two trees is 3 metres and a distance of 2 metres is left from all sides of the boundary of the garden. Find the area of the whole garden?

**Solution**

For columns, there are 15 gaps between the two corner avocado trees $(15 \times 3)$ metres and 2 metres on each side is left. Therefore length is $(45 + 4)m = 49$m.

For rows, there are 13 gaps between the two corner avocado trees $(13 \times 3)$ metres and 2 metres on each side is left. Therefore width is $(39 + 4)m = 43$m.

The area of the garden is length $\times$ width $=49 \times 43 = 2107m^2$.

10. There are 100 questions in a 2 hour examination. Among these questions are 40 mathematics questions. It is suggested that thrice (three times) as much time be spent on each mathematics problem as for each other question. How many minutes should be spent on mathematics problems?

**Solution**

Let $y$ be the total time spent on each mathematics problem. Now, for 40 maths questions, the time taken will be $40y$. For remaining questions, the time taken will be $60 \times \frac{y}{3} = 20y$.

Total time taken to do the examination = 2hrs$= 60 \times 2 = 120$ minutes.

But then $40y + 20y = 120$ implying that $60y = 120 \Longrightarrow y = 2$.

Therefore, the time taken to solve 40 mathematics problems$= 40 \times 2 = 80$ minutes.

## SECTION B:

11. The heights of 4 athletes on a team are 135 cm, 160 cm, 170 cm, and 175 cm. Kiplimo joins the team. On the new team of 5 athletes, the modal height of the players is equal to the median height which is equal to the mean (average) height. How tall is Kiplimo?

**Solution**

Since the heights of the 4 athletes on the team are all different, then if Kiplimo's height is different from each of these, there is no single mode height. Therefore, Kiplimo's height must be equal to the height of one of the 4 athletes on the team for there to be a single mode. If Kiplimo's height is 135 cm, then the median height of the 5 athletes is 160 cm which is not possible, since the median does not equal the mode. Similarly, if Kiplimo's height is 175 cm, then the median height of the 5 athletes is 170cm which is not possible.

Therefore, Kiplimo's height must equal 160 cm or 170 cm, since in either case the median height of the 5 athletes will equal Kiplimo's height, which is the mode.

If Kiplimos's height is 170 cm, then the mean height of the 5 athletes is

$$\frac{135 + 160 + 170 + 170 + 175}{5} = 162cm.$$

If Kiplimo's height is 160 cm, then the mean height of the 5 athletes is

$$\frac{135 + 160 + 160 + 170 + 175}{5} = 160cm.$$

When Kiplimo's height is 160 cm, the heights of the 5 athletes (measured in cm) are: $135, 160, 160, 170$, and 175. In this case, each of the mode, median and mean height of the 5 athletes equals 160 cm.

12. Seven distinct positive integers will be placed in the seven boxes of the "H"-shaped Figure 1 below. The integers must be placed so that the three integers in the left vertical column, those in the right vertical column, and those in the one horizontal row all have the same sum. The integers $k$ and $n$ are each between 4 and 18, inclusive. The figure contains seven different integers and the three sums are equal. Determine all possible values of k.

| | | |
|---|---|---|
| 7 | | $n+6$ |
| 10 | $n$ | 18 |
| $k$ | | 4 |

**Solution**

Since the three sums are equal, we have $7+10+k = 10+n+18 = (n+6)+18+4$ which gives $k+17 = n+28 = n+28$, which means that $k = n+11$. Since $k$ is between 4 and 18, inclusive, and $k = n+11$, then we must have $n \leq 7$. Since $n$ is between 4 and 18, inclusive, then $n \geq 4$ as well. Putting this together, $n$ can equal $4, 5, 6$, or 7, which give values for $k$ of $15, 16, 17$, or 18, respectively, and values of $n+6$ of $10, 11, 12$, or 13, respectively. Since the figure already contains 10 and 18, we cannot have $k = 18$ or $n+6 = 10$, which means that $n \neq 7$ and $n \neq 4$. This means that the possible values of $k$ are 16 and 17.

13. Starting on the 22nd floor of their apartment building, Hassan goes up the stairs and Henry goes up by an elevator. When Hassan starts going up, Henry waits for 2 minutes for the elevator and Hassan goes up from each floor to the next floor in 15 seconds. The elevator goes up from each floor to the next floor in 3 seconds. Hassan and Henry arrive on the $n$th floor at exactly the same time. What is the value of $n$?

**Solution**

From the 22nd floor to the nth floor, Hassan and Henry each go up $n-22$ floors. Since Hassan goes from each floor to the next in 15 seconds, this takes his $15 \times (n-22)$ seconds. Meanwhile, Henry waits for 2 minutes implying 120 seconds. Since Henry goes from each floor to the next in 3 seconds, it takes a total of $120 + 3 \times (n-22)$ seconds to reach the nth floor. Since their travel times are equal, then

$$15 \times (n-22) = 120 + 3 \times (n-22)$$

$$15 \times (n-22) - 3 \times (n-2) = 120$$

$$12 \times (n-22) = 120$$

$$n - 22 = 10$$

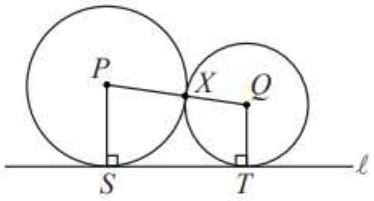
and so $n = 32$.

14. The integers 1 to 32 are spaced evenly and in order around the outside of a circle. Straight lines that pass through the centre of the circle join these numbers in pairs. Which number is paired with 12?

**Solution**

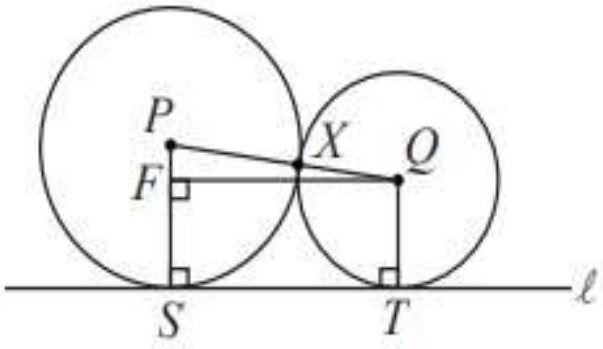
The integers 1 to 32 are spaced evenly and in order around the outside of a circle. Consider drawing a first straight line that passes through the centre of the circle and joins any one pair of these 32 numbers. This leaves 32 - 2 = 30 numbers still to be paired. Since this first line passes through the centre of the circle, it divides the circle in half. In terms of the remaining 30 unpaired numbers, this means that 15 of these numbers lie on each side of the line drawn between the first pair. Let the number that is paired with 12 be $n$. If we draw the line through the centre joining 12 and $n$, then there are 15 numbers that lie between 12 and $n$ (moving in either direction, clockwise or counter clockwise). Beginning at 12 and moving in the direction of the 13, the 15 numbers that lie between 12 and $n$ are the numbers $13, 14, 15, \ldots, 26, 27$. Therefore, the next number after 27 is the number $n$ that is paired with 12. The number paired with 12 is 28.

15. In Figure 2, circles with centres $P$ and $Q$ touch a line $\ell$ (that is, are tangent to $\ell$) at points $S$ and $T$, respectively, and touch each other at point $X$. Because the circles touch at $X$, line segment $PQ$ passes through $X$ and so $PQ = PX + XQ$. Because the circles touch $\ell$ at $S$ and $T$, $PS$ and $QT$ are perpendicular to $\ell$. If the circle with centre $P$ has a radius of 25 and the circle with centre $Q$ has a radius of 16, determine the area of trapezoid $PQTS$.




**Solution**

Draw a perpendicular from $Q$ to $F$ on $PS$ as shown in Figure 2 below. Quadrilateral $FQTS$ has




three right angles and so it must be a rectangle. Therefore, $FS = QT$. Since $QT$ is a radius, then $FS = QT = 16$. Since $PS$ is a radius, then $PS = 25$. Therefore, $PF = PS - FS = 25 - 16 = 9$. We note that $PQ = PX + XQ = 25 + 16 = 41$ (they are both radii). By the Pythagorean Theorem in $\triangle PFQ$, which is right-angled at $F$,

$$FQ = \sqrt{PQ^2 - PF^2} = \sqrt{41^2 - 9^2} = \sqrt{1681 - 81} = 40.$$

Since $FQTS$ is a rectangle, $ST = FQ = 40$. Finally, the area of trapezoid $PQTS$ equals

$$\frac{1}{2}(PS + QT) \times ST = \frac{1}{2}(25 + 16) \times 40 = 41 \times 20 = 820.$$